

VINSUP SKILL ACADEMY

Coimbatore

PROJECT DOCUMENTATION

ANALYZING AND VISUALIZING REGIONAL SALES PERFORMANCE

Domain: Retail Sales Analytics | ABC Electronics

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Domain: Retail Sales Analytics

Dataset: Sales_Performance_Analysis.xlsx (4,000 Rows)

Tools Used: Microsoft Excel

Technical Tags: Excel, Data Cleaning, Pivot Tables, Charts, Dashboard, Data Analysis

1. PROJECT OVERVIEW

Project Title

Analyzing and Visualizing Regional Sales Performance

Organization

ABC Electronics

Domain

Retail Sales Analytics

Tool Used

Microsoft Excel

Dataset

Sales_Performance_Analysis.xlsx (4,000 Rows, 8 Columns)

Problem Statement

ABC Electronics needed to evaluate its sales performance across different regions and product categories. Management required a summarized report with visual insights to identify top-performing regions, product categories, and business trends. The objective was to clean the raw data, perform analysis using Excel, and create an interactive dashboard for effective decision-making.

Business Use Cases

- Identifying top-performing regions and categories.
- Understanding the impact of discounts on sales and profit.
- Supporting inventory and marketing decisions.
- Monitoring key sales metrics efficiently.

Project Deliverables

- Cleaned and standardized dataset.
- Pivot Table analysis.
- Bar Chart, Pie Chart, and Stacked Bar Chart.
- Interactive Dashboard with slicers.
- Regression Analysis.
- Conditional Formatting insights.
- Project Documentation.

2. DATASET DESCRIPTION

The dataset contains 4,000 sales transaction records from November 2022 to November 2024 across four regions.

Column Name	Description
Order ID	Unique Order Identifier
Order Date	Date of Order
Region	North, South, East, West
Product Category	Electronics, Clothing, Furniture, Groceries, Sports
Sales Amount	Revenue Generated
Quantity Sold	Number of Units Sold
Discount (%)	Discount Applied
Profit	Profit Earned

3. DATA CLEANING AND PREPROCESSING

The raw dataset contained inconsistent values, special characters, extra spaces, and duplicate records.

Cleaning Activities Performed

Step 1: Text Standardization

Functions Used:

=TRIM()
=UPPER()
=LOWER()

These functions removed extra spaces and standardized text formatting.

Step 2: Region Correction

Incorrect region names such as:

\$0uTH
EaA\$T
norTh

were converted into:

North
South
East
West

Step 3: Product Category Correction

Incorrect category names were standardized into:

- Electronics
- Clothing
- Furniture
- Groceries
- Sports

Step 4: Handling Missing Values

Missing values were identified and excluded from calculations where necessary.

Step 5: Removing Duplicate Records

Duplicate Order IDs were identified and removed to maintain data accuracy.

Final Outcome

The cleaned dataset contained only valid and standardized records suitable for analysis.

4. EXCEL FORMULAS USED

Total Sales by Region

Formula:

```
=SUMIF(C:C,"North",E:E)
```

This formula was used to calculate sales for each region.

Average Discount for Furniture

Formula:

```
=AVERAGEIF(D:D,"Furniture",G:G)
```

Average Profit for Furniture

Formula:

```
=AVERAGEIF(D:D,"Furniture",H:H)
```

Average Sales by Region

Formula:

```
=AVERAGEIF(C:C,"North",E:E)
```

Data Merge Using VLOOKUP

Formula:

```
=VLOOKUP(C2,RegionAvgTable,2,0)
```

This formula was used to merge average regional sales into the main dataset.

5. PIVOT TABLE ANALYSIS

Pivot Tables were created to summarize sales and profit information.

Configuration

Rows: • Region

Columns: • Product Category

Values: • Sum of Sales Amount • Sum of Profit

Slicers: • Region • Product Category

Benefits

- Quick summary of large datasets.
 - Interactive filtering.
 - Better decision-making.
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6. CHARTS AND VISUALIZATIONS

Bar Chart – Total Sales by Region

Purpose:

To compare regional sales performance.

Insight:

North region generated the highest sales.

Pie Chart – Product Category Contribution

Purpose:

To display percentage contribution of each product category.

Insight:

Clothing contributed the highest percentage of total sales.

Stacked Bar Chart – Region vs Product Category Sales

Purpose:

To compare category-wise sales across different regions.

Insight:

North region performed consistently across all product categories.

7. INTERACTIVE DASHBOARD

An interactive Excel dashboard was created to summarize key metrics.

Dashboard Components

KPI Cards

- Total Sales
- Total Profit
- Average Discount
- Total Orders

Visualizations

- Bar Chart
- Pie Chart
- Stacked Bar Chart

Interactive Features

- Region Slicer
- Product Category Slicer
- Date Filter

Dashboard Benefits

- Real-time filtering.
 - Easy interpretation of business performance.
 - Quick decision-making support.
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8. REGRESSION ANALYSIS

Regression analysis was performed to understand the relationship between Discount (%) and Sales Amount.

Method Used

- Scatter Plot
- Trendline

Purpose

To determine whether discounts influence sales performance.

Observation

The analysis showed that discount percentage had very little impact on sales revenue, indicating that discount strategies should be reviewed carefully.

9. CONDITIONAL FORMATTING

Conditional formatting was applied to highlight important transactions.

Rules Used

- Profit Margin > 50%
- Sales Amount > ₹4,000

Benefits

- Quickly identifies high-performing orders.
 - Improves data visibility.
 - Supports faster business analysis.
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10. KEY BUSINESS INSIGHTS

Regional Performance

- North region achieved the highest sales performance.
- South region recorded the lowest sales performance.

Product Category Performance

- Clothing was the top-performing category.
- Sports contributed the lowest share of sales.

Discount Impact

- Discounts had minimal influence on overall sales performance.

Profitability

- The business maintained a strong overall profit margin.
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11. CONCLUSION

This project successfully demonstrated the complete data analysis process using Microsoft Excel. Data cleaning, formula-based calculations, Pivot Tables, visualizations, dashboard creation, and regression analysis were used to transform raw data into meaningful business insights.

The analysis revealed that the North region generated the highest sales, Clothing was the leading product category, and discounts had limited impact on revenue generation. The interactive dashboard provides management with a powerful tool for monitoring performance and supporting strategic decision-making.

Overall, the project highlights how Microsoft Excel can be effectively used for business analytics, reporting, and visualization.

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